Module 4 – Exercise 1: Guess a Number Between 1 and 100

Steven Francis · SDEV120 · Flowchart built in Lucidchart

The task

Draw a structured flowchart of the process of guessing a number between 1 and 100. After each guess the player is told whether the guess is too high or too low, and the process continues until the player guesses the correct number.

How the flowchart works

- secretNumber is set with random(MIN, MAX). guess starts at 0, which can never equal the secret number, so the main loop always runs at least once.
- Main loop: while guess <> secretNumber, prompt for and input a guess.
- Input validation loop: while the guess is below 1 or above 100, display "Invalid – must be 1 to 100" and input the guess again.
- attempts is increased by 1 for every valid guess.
- Nested selection: if guess > secretNumber display "Too high"; otherwise, if guess < secretNumber display "Too low". If neither is true the guess is correct, and the loop test ends the loop.
- After the loop: display "Correct! You guessed it in N tries", then stop.

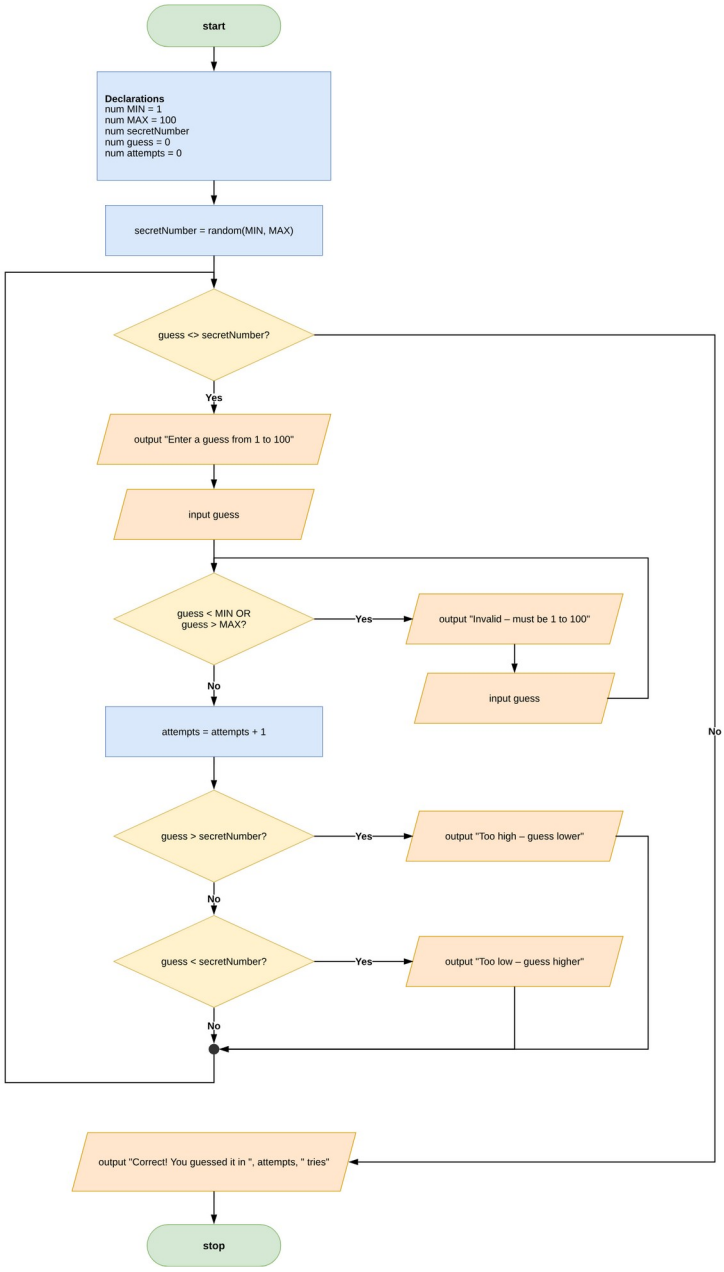
Structures used

The flowchart is structured: it uses only the three basic structures, and each structure has one entry and one exit.

- **Sequence** – steps run one after another, top to bottom.
- **Selection** – a diamond tests a Boolean expression and takes exactly one Yes/No path; both paths rejoin at a single point (the small black dot).
- **Loop** – the test sits at the top of the loop; while it is true the body repeats, then control returns to the same test.

Symbol key: green rounded = start/stop, blue rectangle = process, orange parallelogram = input/output, yellow diamond = decision.

Flowchart



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